Boundary convergence of planar Burg and matrix entropy mirror descent

Abstract

We study boundary convergence and selection for exact fixed-step mirror descent. For polynomial objectives that may be nonconvex, Burg descent on a planar affine simplex slice converges to one stationary point, while radial binary entropy descent on any finite-dimensional ball has finite Euclidean length. The latter is a finite-length theorem for all complex Hermitian two-by-two density matrices. For polynomials that are convex on the entire Hermitian matrix space, in arbitrary finite matrix size, we prove convergence to the unique initial-relative-entropy minimizer among the constrained loss minimizers. The result allows singular solutions and includes inconsistent quadratic data. The proofs retain the actual discrete recurrence: tangential dual coordinates handle polygon edges, a sphere-gradient dissipation estimate handles entropy balls, and a polynomial estimate controls accumulated gradients to prove matrix selection without boundary reciprocity. Explicit trajectories exhibit nonisolated boundary limits, noncommuting updates and rank-deficient selected states.

Keywords: Burg entropy; matrix entropy; mirror descent; boundary convergence; polynomial optimization

2020 Mathematics Subject Classification. Primary 65K10; Secondary 90C23, 90C25.

1 The planar convergence theorem

Let $n \geq 3$ and write

$$\Delta_n = \{x \in \mathbb{R}^n : x_i \geq 0 \ (1 \leq i \leq n), \ \sum_i x_i = 1\}. \tag{1}$$

Let A be a two-dimensional affine plane contained in $\{x : \sum_i x_i = 1\}$, and suppose that A contains a strictly positive point. Then

$$P = A \cap \Delta_n, \qquad P_+ = A \cap \mathbb{R}^n_{++} = \operatorname{ri} P \tag{2}$$

is a compact convex polygon with nonempty relative interior. The number n of ambient coordinates is unrestricted; the feasible affine dimension is exactly two.

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a real polynomial. Choose any $L > 0$ such that

$$\|\nabla f(x) - \nabla f(y)\|_2 \leq L\|x - y\|_2 \qquad (x, y \in P). \tag{3}$$

Such a constant exists by compactness. Fix $0 < \alpha < 1/L$, let

$$h(x) = -\sum_{i=1}^{n} \log x_i, \qquad D_h(y, x) = h(y) - h(x) - \langle \nabla h(x), y - x \rangle, \tag{4}$$

and start from any $x_0 \in P_+$. We study the exact Burg mirror iteration

$$x_{k+1} = \operatorname*{arg\,min}_{y \in P_+} \left\{ \langle \nabla f(x_k), y - x_k \rangle + \frac{1}{\alpha} D_h(y, x_k) \right\}, \qquad k \geq 0. \tag{5}$$

Theorem 1. *Every update* (5) *has a unique minimizer in* P_+*. The objective values are nonincreasing, and*

$$\sum_{k=0}^{\infty} \|x_{k+1} - x_k\|_2^2 < \infty. \tag{6}$$

The whole sequence converges to one point $\bar{x} \in P$*, which satisfies*

$$\langle \nabla f(\bar{x}), y - \bar{x} \rangle \geq 0 \qquad \textit{for every } y \in P. \tag{7}$$

Thus $\bar{x}$ *is a KKT point for the affine and nonnegativity constraints. Convexity of* f*, isolated stationary points, strict complementarity and an interior limit are not required.*

The theorem concerns one stationary point of the nonconvex problem; it does not assert global optimality. It gives no convergence rate or finite-length conclusion for a boundary-convergent trajectory. Higher-dimensional feasible slices are outside its scope. In particular, allowing more than three ambient coordinates does not remove the planar hypothesis.

1.1 Relation to existing boundary results

Classical Bregman analysis is delicate for the Burg kernel because h and $D_h(\bar{x}, x)$ become infinite at a boundary point $\bar{x}$. Chen, Fan and So [1, Section 6] identify open-domain kernels as a remaining direction in their iterate-convergence framework. Their closed-domain results do not supply the Burg boundary case. Ding and Toh [2, Definition 3.1 and Theorem 4.1] use a metric-dependent transformation; for Burg its coordinates are $\log x_i$. A bounded trajectory in P can therefore have unbounded transformed coordinates. The example in Section 6 exhibits this explicitly.

Convergence guarantees for logarithmic barriers are nevertheless already available in important settings. De Marchi, Malitsky and Taylor [3, Assumption 1 and Theorem 2] prove objective-gap bounds of order $\log k/k$ for convex mirror descent with logarithmic barriers, including boundary minimizers and affine constraints. The present result concerns whole-iterate convergence for possibly nonconvex polynomial objectives in feasible dimension two. It is a restricted instance of the open-domain direction, not a resolution for arbitrary kernels or dimensions.

The interior part of our proof uses the classical analytic gradient inequality, in the form recalled by Absil, Mahony and Andrews [4, Lemma 2.1]. The boundary part uses the original discrete mirror optimality equation and polynomial edge structure. General planar Bregman dynamics need not converge: the counterexample of Bolte and Pauwels [5,

Section 5.7] employs a specially constructed Legendre kernel. The fixed logarithmic kernel here is essential to the regular tangential coordinate used below. Comparisons with the 2026 works refer to the public versions specified in the references.

The second setting, developed in Section 7, uses a nonseparable radial entropy kernel on a ball of arbitrary fixed finite dimension. An exact two-state estimate gives finite trajectory length, including at nonisolated boundary limits. In dimension three this is the matrix-exponential algorithm for all complex Hermitian two-by-two density matrices. It addresses this particular nonseparable-kernel setting identified as a remaining direction by Ding and Toh [2, Section 6]. Its nonconvex matrix consequence is confined to two-by-two states. Section 8 treats arbitrary matrix sizes for polynomials convex on the entire ambient Hermitian space and identifies the initial-relative-entropy selected minimizer. Inconsistent quadratic data are included. Neither result concerns arbitrary nonseparable kernels.

2 Descent and capture by an interior cluster point

Let $V \in \mathbb{R}^{n\times 2}$ have orthonormal columns spanning the direction space of A. Write $g_k = \nabla f(x_k)$ and $d_k = x_{k+1} - x_k$.

Lemma 1. *The iteration is well defined and obeys*

$$V^T\big(\nabla h(x_{k+1}) - \nabla h(x_k)\big) = -\alpha V^T g_k. \tag{8}$$

Moreover, with $\sigma = \alpha^{-1} - L/2 > 0$,

$$f(x_k) - f(x_{k+1}) \geq \frac{1}{\alpha}\sum_i \frac{d_{k,i}^2}{x_{k,i}x_{k+1,i}} - \frac{L}{2}\|d_k\|_2^2 \geq \sigma\|d_k\|_2^2. \tag{9}$$

The cluster set S is nonempty, compact and connected. There is a number ℓ such that $f(x_k) \downarrow \ell$, $f = \ell$ on S, $d_k \to 0$ and $\operatorname{dist}(x_k, S) \to 0$.

Proof. For fixed x_k, the subproblem is a strictly convex function of y on P_+. Its linear part stays bounded on P, whereas $h(y) \to +\infty$ at every relative boundary point. Its finite sublevels are compact subsets of P_+, so its unique minimizer exists. The affine first-order condition is (8).

Since d_k belongs to the direction space of A, taking its inner product with the unprojected difference gives

$$\langle g_k, d_k\rangle = -\frac{1}{\alpha}\langle \nabla h(x_{k+1}) - \nabla h(x_k), d_k\rangle = -\frac{1}{\alpha}\sum_i \frac{d_{k,i}^2}{x_{k,i}x_{k+1,i}}. \tag{10}$$

The descent inequality following from (3) applies on the segment in P between the consecutive iterates. Combining it with (10) proves the first inequality in (9). The second uses $0 < x_{k,i} \leq 1$.

Compactness bounds f below and therefore gives square-step summability, $d_k \to 0$, and the asserted value and cluster properties except connectedness. If S were disconnected, two nonempty disjoint compact pieces would have positive distance. The tail would lie in their disjoint small neighborhoods and visit each infinitely often. A transition between them with vanishing steps would produce an additional cluster point outside both neighborhoods, a contradiction. ∎

Lemma 2. *If $S \cap P_+ \neq \varnothing$, then the whole sequence converges to a point in P_+.*

Proof. Let $q \in S \cap P_+$. In a fixed affine ball $B_A(q, 2r) \subset P_+$ the restricted Hessian of h has a finite upper bound M. Whenever the endpoints of a step lie in this ball, (8) implies

$$\|V^T g_k\|_2 \leq \frac{M}{\alpha} \|d_k\|_2. \tag{11}$$

A subsequence approaching q has its successors approaching q too. Thus $V^T \nabla f(q) = 0$.

The local analytic gradient inequality for f restricted to A supplies a concave function φ, continuous at zero with $\varphi(0) = 0$, such that, after decreasing r,

$$\varphi'(f(x) - \ell) \|V^T \nabla f(x)\|_2 \geq 1 \tag{12}$$

near q whenever $0 < f(x) - \ell$ is sufficiently small. If some iterate has value ℓ, (9) makes every subsequent step zero. Otherwise put $e_k = f(x_k) - \ell > 0$. Concavity, (9), (11) and (12) give

$$\|d_k\|_2 \leq \frac{M}{\alpha\sigma} \big(\varphi(e_k) - \varphi(e_{k+1})\big) \tag{13}$$

for steps starting in $B_A(q, r)$ with endpoints in the larger ball.

Choose a sufficiently late visit k_0 with $\|x_{k_0} - q\|_2 < r/4$, all subsequent steps shorter than $r/4$, and $M\varphi(e_{k_0})/(\alpha\sigma) < r/4$. Up to a first exit from $B_A(q, r)$, every step remains in $B_A(q, 2r)$ and (13) bounds the total travel by $r/4$. Such an exit is impossible. The infinite telescope now gives a Cauchy tail. Since q is a cluster point, its limit is q. ∎

3 A regular dual coordinate at each open edge

Suppose henceforth that the sequence does not converge. Lemma 2 places the entire cluster set on the polygon boundary $\Gamma = \partial_A P$; in particular

$$\operatorname{dist}(x_k, \Gamma) \longrightarrow 0. \tag{14}$$

Fix an open edge E and choose orthonormal affine coordinates (u, v) on A, with $u = 0$ on its supporting line, $u > 0$ inside P, and v tangential to E. Write

$$x_i(u, v) = a_i u + b_i v + c_i, \qquad F(u, v) = f(x(u, v)), \qquad H(u, v) = h(x(u, v)). \tag{15}$$

If x_i vanishes on E, its affine formula has $b_i = c_i = 0$ and $a_i > 0$. Every other coordinate is positive on any compact subinterval of E. Hence the tangential dual coordinate

$$T(u,v) = \partial_v H(u,v) = -\sum_{i:\, x_i|_E \not\equiv 0} \frac{b_i}{x_i(u,v)} \tag{16}$$

extends smoothly to a strip adjoining that subinterval. The singular terms $-\log(a_i u)$ have zero tangential derivative.

Orthonormality gives $\sum_i b_i^2 = 1$, while the coordinates vanishing on E have $b_i = 0$. Thus, at the edge and inside the strip,

$$\partial_v T(u,v) = \sum_{i:\, x_i|_E \not\equiv 0} \frac{b_i^2}{x_i(u,v)^2} > 0. \tag{17}$$

At $u = 0$ this derivative is at least one. Each level $T = T(0, v_0)$ near an open-edge point is consequently a graph $v = \psi(u)$ crossing the narrow strip. Pairing (8) with the fixed tangential direction yields the exact difference equation

$$T(u_{k+1}, v_{k+1}) - T(u_k, v_k) = -\alpha\, \partial_v F(u_k, v_k). \tag{18}$$

The affine chart and the tangential direction can be continued over A, so this identity holds along the entire orbit; smooth extension to the boundary is used only within the selected edge strip.

Lemma 3. *Suppose $\partial_v F$ has one strict sign on a strip $0 < u < \varepsilon$, $v \in I$ around an open-edge point $(0, v_0)$. Then a sufficiently late crossing of the nearby level $T = T(0, v_0)$ can occur only in the direction prescribed by* (18)*. No positive lower bound on the increment's magnitude is required.*

Proof. Choose nested neighborhoods of the level graph within the strip. The function T and its derivatives are bounded on the closure of the larger neighborhood. Because $d_k \to 0$, any sufficiently late change of side has its starting point in that neighborhood and both endpoints close to the graph. The sign of (18) excludes one direction of crossing, including a step starting exactly on the level. The strip's interior cap $u = \varepsilon$ has positive distance from Γ after restricting v to a compact edge interval. By (14), a late passage cannot bypass the graph through that cap. Vanishing steps also exclude a jump over a fixed-width neighborhood. These assertions concern consecutive discrete iterates and retain the u-dependence of the level graph. ∎

Suppose now that $f = \ell$ on an edge interval. Since F is polynomial, $F(0, v) = \ell$ on the whole supporting line, and it has a finite expansion

$$F(u,v) = \ell + \sum_{j=1}^{D} u^j A_j(v). \tag{19}$$

If all A_j are constant, then $\partial_v F \equiv 0$ and (18) makes $T(x_k)$ constant along the orbit. Its strictly increasing edge trace prevents an interval of E from being contained in S.

Otherwise let m be the smallest index for which A_m is nonconstant. Then

$$\partial_v F(u,v) = u^m\big(A'_m(v) + uR(u,v)\big) \tag{20}$$

for a polynomial R. At every v_0 where $A'_m(v_0) \neq 0$, this derivative has a uniform strict sign on a sufficiently narrow strip. Indeed, first choose a compact interval on which $|A'_m| \geq c > 0$, and then choose ε so that $\varepsilon \sup|R| < c/2$. Lower-order coefficients depending only on u vanish from (20); they need not be zero.

4 Excluding nontrivial boundary cluster sets

We now use planarity. A proper compact connected subset of the polygon boundary is either one point or a closed arc. The entire perimeter requires a separate argument.

4.1 A proper boundary arc

Suppose that S is a nontrivial proper arc. It contains an open interval of an edge E. The expansion (19) applies there. The case $\partial_v F \equiv 0$ has already been excluded by the invariant T. Otherwise, the nonzero polynomial A'_m has only finitely many roots. Choose v_0 strictly inside the cluster interval where $A'_m(v_0) \neq 0$. Equation (20) and Lemma 3 provide a one-direction crossing barrier at

$$T(u,v) = t_0 := T(0,v_0). \tag{21}$$

There are cluster points on both sides of this wall.

We describe why the orbit cannot return by another route. Choose $c \in P_+$. Near Γ, write each point uniquely as

$$x = c + (1-\rho)(p-c), \qquad p \in \Gamma, \quad 0 \leq \rho < \rho_0 < 1. \tag{22}$$

The maps $x \mapsto (p,\rho)$ and their inverse are continuous and uniformly continuous on a fixed closed collar. This follows by intersecting the ray from c with the boundary of the convex polygon. Condition (14) implies $\rho_k \to 0$. Both p_k and x_k have cluster set S, and the adjacent collar coordinates differ by quantities tending to zero.

Cut the perimeter at a closed subarc in a gap disjoint from S. The tail avoids that cut and admits a continuous perimeter coordinate $s(p)$ on the remaining arc. Orient s to increase with v near v_0. In the collar, the wall (21) has the form

$$s(p) = s_0(\rho), \qquad s_0(\rho) \longrightarrow s(p_0), \quad p_0 = x(0,v_0). \tag{23}$$

Indeed, for each fixed ρ, the normal coordinate u is constant as p varies along this edge, and (17) gives strict monotonicity in its tangential coordinate. The implicit function theorem gives the continuous function s_0.

The sequence $w_k = s(p_k) - s_0(\rho_k)$ has both a strictly positive and a strictly negative cluster value. It must change signs in both directions arbitrarily late. Its increments tend to zero, so every sufficiently late sign change starts near the local wall. There the sign of w_k agrees with the sign of $T(x_k) - t_0$. One direction is prohibited by (18). This contradiction excludes a nontrivial proper arc. The perimeter cut prevents a cyclic bypass, and the collar description retains all normal displacement.

4.2 The whole polygon perimeter

Suppose instead that $S = \Gamma$. Then $f = \ell$ on every edge. The restriction of $f - \ell$ to the affine plane is divisible by the product of the distinct affine equations of all supporting edge lines: vanishing on an edge interval gives vanishing on its whole line, and the distinct linear factors are relatively prime.

If this restriction is identically zero, its tangent gradient is zero everywhere and (5) is the identity. Thus assume that it is nonzero. Fix one edge with normal coordinate u. Let $m \geq 1$ be the smallest power of u occurring in the restriction. Then

$$F(u, v) - \ell = u^m \big(B(v) + uQ(u, v)\big), \qquad B \not\equiv 0, \tag{24}$$

where B and Q are polynomials. The factors corresponding to both adjacent edge lines are coprime with u and still divide the coefficient at $u = 0$. Consequently, if $a < b$ are the tangential endpoints of the selected edge,

$$B(a) = B(b) = 0. \tag{25}$$

Moreover $B(v) \geq 0$ for every $v \in (a, b)$. To see this, choose a subsequence approaching the corresponding edge point, which exists because $S = \Gamma$. Divide $f(x_k) - \ell \geq 0$ by $u_k^m > 0$ and pass to the limit in (24). Since B is nonzero and nonnegative with zero endpoint values, the mean value theorem supplies

$$a < p < q < b, \qquad B'(p) > 0, \qquad B'(q) < 0. \tag{26}$$

Bounding the polynomial remainder as in (20) gives $\partial_v F > 0$ near $(0, p)$ and $\partial_v F < 0$ near $(0, q)$ for every sufficiently small positive u.

Set $t_p = T(0, p) < t_q = T(0, q)$. Choose $\varepsilon > 0$ small enough that

$$U = \{x \in P_+ : 0 < u < \varepsilon,\ t_p < T(u, v) < t_q\} \tag{27}$$

lies in the selected open-edge strip, with the two lateral walls in the respective sign neighborhoods. Equation (18) points outward at both walls. At $T = t_p$, an entry from

the left would require T to increase, whereas its increment is negative. At $T = t_q$, an entry from the right would require T to decrease, whereas its increment is positive.

There is no late entry from outside U to inside U. A crossing of either lateral wall starts in its sign neighborhood once d_k is sufficiently small, and has just been excluded, including exact contact with a wall. Entry through the top $u = \varepsilon$ is excluded by (14), because this compact cap stays away from every edge. Entry through $u = 0$ is impossible since all finite iterates belong to P_+. Vanishing steps prevent jumping over one of these fixed neighborhoods.

But full-boundary accumulation requires infinitely many visits both to U and to points a fixed distance from its closure. Such visits require entries arbitrarily late. The contradiction excludes $S = \Gamma$.

The connected cluster set must therefore be a singleton unless Lemma 2 has already proved convergence. Compactness shows that the entire sequence converges in either case.

5 Stationarity of the limiting point

Let $x_k \to \bar{x} \in P$. Fix any $y \in P_+$ and put $w = y - \bar{x}$, which belongs to the direction space of A. Summing the exact dual identity (8) and pairing with w gives

$$\langle \nabla h(x_K) - \nabla h(x_0), w \rangle = -\alpha \sum_{k=0}^{K-1} \langle \nabla f(x_k), w \rangle. \tag{28}$$

For each index with $\bar{x}_i > 0$, the corresponding kernel-gradient term remains bounded. For $\bar{x}_i = 0$, one has $w_i = y_i > 0$ and

$$w_i \, \partial_i h(x_K) = -\frac{y_i}{x_{K,i}} < 0. \tag{29}$$

Hence the left side of (28) is bounded above by a constant independent of K. Dividing by K and using continuity of ∇f gives $\langle \nabla f(\bar{x}), y - \bar{x} \rangle \geq 0$.

For arbitrary $y \in P$, mix y with a fixed point of P_+ and let the mixing coefficient tend to zero. The inequality extends by continuity to every $y \in P$, proving (7). This argument does not require bounded affine multipliers, reciprocal coordinates or logarithmic coordinates. Together with the preceding sections it completes the proof of Theorem 1.

6 A nonisolated boundary limit

The theorem includes trajectories to which a bounded-log-coordinate result cannot be applied.

Example 1. On Δ_3, let

$$f(x) = x_1 x_2 x_3, \qquad \alpha = \tfrac{1}{2}, \qquad x_0 = (\tfrac{1}{4}, \tfrac{3}{8}, \tfrac{3}{8}). \tag{30}$$

Then (5) converges to $(0, \frac{1}{2}, \frac{1}{2})$, with $x_{k,1} \downarrow 0$. Every point of the boundary is a global minimizer and a KKT point, so the limiting stationary point is nonisolated.

Proof. The Hessian of f is symmetric, has zero diagonal, and its absolute row sums on Δ_3 are $1 - x_i \le 1$. Thus $\|\nabla^2 f\|_2 \le 1$ along every segment in the simplex, and $L = 1$ is valid in (3).

Symmetry and uniqueness of the mirror subproblem preserve $x_{k,2} = x_{k,3}$. Write

$$x_k = (u_k, (1-u_k)/2, (1-u_k)/2), \qquad R(u) = \frac{1}{u} - \frac{2}{1-u}. \tag{31}$$

Subtracting the first and second coordinates of the mirror optimality equation cancels the equality multiplier and gives

$$R(u_{k+1}) - R(u_k) = \frac{\alpha}{4}(1-u_k)(1-3u_k). \tag{32}$$

The function R is strictly decreasing on $(0, 1)$. Starting from $u_0 = 1/4$, the positive right side implies $0 < u_{k+1} < u_k \le 1/4$. If its decreasing limit were positive, the right side would tend to a positive number while the left side would tend to zero. Hence $u_k \to 0$.

The polynomial is nonnegative on the simplex and vanishes on its boundary. At an open-edge point, its two positive-coordinate derivatives are zero and its remaining derivative is nonnegative; at a vertex all derivatives vanish. These are the simplex KKT conditions. Finally $\log x_{k,1} \to -\infty$ along this actual trajectory. ∎

7 Entropy mirror descent on finite-dimensional balls

A different exact dissipation estimate treats a nonseparable radial entropy kernel in every finite dimension. At dimension three, it gives the matrix-exponentiated-gradient update [6] on all complex Hermitian two-by-two density matrices.

Fix a finite integer $d \ge 2$ and let $B = \{r \in \mathbb{R}^d : \|r\|_2 \le 1\}$. For $0 \le \rho < 1$, define

$$\psi(\rho) = \frac{1+\rho}{2}\log\frac{1+\rho}{2} + \frac{1-\rho}{2}\log\frac{1-\rho}{2}, \qquad h_E(r) = \psi(\|r\|_2). \tag{33}$$

Let $F : \mathbb{R}^d \to \mathbb{R}$ be a real polynomial, and choose $L_E > 0$ such that ∇F is L_E-Lipschitz on B. Fix $0 < \alpha < 1/L_E$ and $r_0 \in \operatorname{int} B$. We consider

$$r_{k+1} = \operatorname*{arg\,min}_{s \in \operatorname{int} B} \left\{ \langle \nabla F(r_k), s - r_k \rangle + \frac{1}{\alpha} D_{h_E}(s, r_k) \right\}. \tag{34}$$

The constant L_E uses Euclidean ball coordinates; its matrix counterpart is specified below.

Theorem 2. *Every update* (34) *has a unique interior minimizer. The values $F(r_k)$ are nonincreasing and the whole trajectory has finite Euclidean length:*

$$\sum_{k=0}^{\infty} \|r_{k+1} - r_k\|_2 < \infty. \tag{35}$$

It converges to one $\bar{r} \in B$ satisfying

$$\langle \nabla F(\bar{r}), s - \bar{r} \rangle \geq 0 \qquad (s \in B). \tag{36}$$

The conclusion allows nonconvex objectives, boundary limits and nonisolated stationary sets. Bounded dual coordinates, a uniform interior margin, strict complementarity and an eventual direction are not assumed. No quantitative rate is asserted.

7.1 Exact descent and a compact analytic lift

For $\rho = \|r\|_2 > 0$, differentiation gives

$$q = \nabla h_E(r) = \operatorname{atanh}(\rho)\frac{r}{\rho}, \qquad Q(q) = \frac{\tanh\|q\|_2}{\|q\|_2} q = r. \tag{37}$$

Both maps have their continuous values at zero. The radial and tangential eigenvalues of $\nabla^2 h_E$ are $1/(1-\rho^2)$ and $\operatorname{atanh}(\rho)/\rho$, respectively; at zero both are one. Thus $\nabla^2 h_E \succeq I$, and its gradient is a bijection onto $\mathbb{R}^d$. The unique stationary point of the strongly convex subproblem is its global minimizer, and its exact recurrence is

$$q_{k+1} = q_k - \alpha g_k, \qquad g_k = \nabla F(r_k). \tag{38}$$

Put $d_k = r_{k+1} - r_k$ and $\mathcal{S}_k = \langle q_{k+1} - q_k, d_k \rangle$. Strong monotonicity gives $\mathcal{S}_k \geq \|d_k\|_2^2$, so smoothness on the actual segment yields

$$F(r_k) - F(r_{k+1}) \geq \frac{\mathcal{S}_k}{\alpha} - \frac{L_E}{2}\|d_k\|_2^2 \geq \kappa \mathcal{S}_k, \qquad \kappa = \alpha^{-1} - L_E/2 > 0. \tag{39}$$

In particular, $F(r_k) \downarrow \ell$ and the squared steps are summable. A zero step forces $g_k = 0$ and makes the entire tail constant.

Consider the compact sphere lift and its analytic objective

$$z(r) = (r, \sqrt{1 - \|r\|_2^2}) \in \mathbb{S}^d, \qquad \mathcal{F}(x, s) = F(x)\big|_{\mathbb{S}^d}. \tag{40}$$

Projection of $(\nabla F(r), 0)$ onto the sphere's tangent space gives

$$W(r) := \|\operatorname{grad}_{\mathbb{S}^d} \mathcal{F}(z(r))\|_2 = \sqrt{\|\nabla F(r)\|_2^2 - \langle r, \nabla F(r)\rangle^2}. \tag{41}$$

If $r = \rho e$ and $\nabla F(r) = ae + b$ with $b \perp e$, then $W(r)^2 = (1-\rho^2)a^2 + \|b\|_2^2$. The target length remains the physical length in (35); the lifted updates are not being identified with gradient steps.

7.2 A uniform two-state dissipation estimate

Let

$$\begin{aligned} M = \alpha \max_B \|\nabla F\|_2, \quad T = \max\{2, 2M\}, \quad E = e^{2M}, \\ c_0 = \min\{E^{-1}, \tfrac{1}{2}\tanh 1\}, \qquad c = \min\{\operatorname{sech}^2(T+M), c_0/(5E)\} > 0. \end{aligned} \tag{42}$$

We claim that every actual update satisfies

$$\mathcal{S}_k \geq c\alpha W(r_k)\|d_k\|_2. \tag{43}$$

All constants depend only on the fixed problem and step, not on the index or a boundary margin.

Write $q = q_k$, $q' = q_{k+1}$ and $t = \|q\|_2$. The dual jump has norm at most M. If $t \leq T$, its straight segment lies in the ball of radius $T + M$. The radial and tangential eigenvalues of DQ are $\operatorname{sech}^2 t$ and $\tanh(t)/t$, with continuous value one at zero. The latter is at least the former, and both are at most one. Integration on the segment gives

$$\mathcal{S}_k \geq \operatorname{sech}^2(T + M)\|q' - q\|_2^2, \qquad \|d_k\|_2 \leq \|q' - q\|_2. \tag{44}$$

Since $\alpha W(r_k) \leq \alpha\|g_k\|_2 = \|q' - q\|_2$, this proves (43) in this case, including zero dual vectors.

Now let $t \geq T$ and write

$$\begin{gathered} q = te, \quad q' = t'e', \quad \|e\|_2 = \|e'\|_2 = 1, \quad \rho = \tanh t, \quad \rho' = \tanh t', \\ u = |t' - t|, \qquad D = \|e' - e\|_2, \qquad \delta = \operatorname{sech}^2 t. \end{gathered} \tag{45}$$

The bounded jump implies $u \leq M$, $t/2 \leq t' \leq 3t/2$ and $t' \geq 1$. The absolute logarithmic derivative of $\operatorname{sech}^2 s$ is at most two. On the radial interval between t and t' this gives

$$|\rho' - \rho| \leq E\delta u, \qquad (t' - t)(\rho' - \rho) \geq E^{-1}\delta u^2. \tag{46}$$

Expanding the actual two-state inner product yields

$$\mathcal{S}_k = (t' - t)(\rho' - \rho) + \tfrac{1}{2}(t\rho' + t'\rho)D^2 \geq c_0(\delta u^2 + tD^2), \tag{47}$$

whereas

$$\|d_k\|_2 \leq E\delta u + D. \tag{48}$$

These estimates allow the radial increment to have either sign.

Decompose $g_k = ae + b$, $b \perp e$. From the exact dual difference,

$$\alpha a = t - t'\langle e, e'\rangle, \qquad \alpha\|b\|_2 = t'\|e' - \langle e, e'\rangle e\|_2. \tag{49}$$

Using $1 - \langle e, e'\rangle = D^2/2$, $D \leq 2$ and the bound of the sine by the chord gives

$$\alpha W(r_k) \leq \sqrt{\delta}(u + t'D^2/2) + t'D \leq \sqrt{\delta}u + 3tD. \tag{50}$$

The elementary inequality $t\sqrt{\delta} = t\operatorname{sech} t \leq 1$ follows from $\cosh t \geq t$. Multiply (48) and (50), then use $2\sqrt{\delta}uD \leq \delta u^2 + D^2$:

$$\begin{aligned} \alpha W(r_k)\|d_k\|_2 &\leq E\delta^{3/2}u^2 + 3tD^2 + (\sqrt{\delta} + 3Et\delta)uD \\ &\leq 3E\delta u^2 + 5EtD^2 \leq 5E(\delta u^2 + tD^2). \end{aligned} \tag{51}$$

Indeed, the intermediate coefficients are bounded by $E + (1 + 3E)/2 \leq 3E$ and $3 + (1 + 3E)/2 \leq 5E$, since $E, t \geq 1$. Equations (47) and (51) prove (43). This is a finite-step calculation in arbitrary tangential dimension, including simultaneous radial and tangential motion.

7.3 Finite length and the limiting KKT sign

Combining (39) and (43) gives

$$F(r_k) - F(r_{k+1}) \geq \gamma W(r_k)\|d_k\|_2, \qquad \gamma = (1 - \alpha L_E/2)c > 0. \tag{52}$$

Let K be the cluster set of the compact lifted sequence $z(r_k)$. It is nonempty and compact, and $\mathcal{F} = \ell$ on K. In a regular analytic sphere chart at each point of K, the local analytic gradient inequality [4, Lemma 2.1] applies to $\mathcal{F}$. The chart derivative is bounded on a smaller neighborhood, so its gradient is bounded by a constant times the intrinsic gradient. Noncritical points satisfy the same type of inequality by a positive gradient lower bound; stationarity of K is not presupposed.

A finite cover of K, the maximum of its finitely many exponents, and a value gap smaller than one give constants $C > 0$, $0 \leq \theta < 1$ such that, on the whole sufficiently late tail,

$$(F(r_k) - \ell)^\theta \leq CW(r_k) \qquad \text{whenever } F(r_k) > \ell. \tag{53}$$

Here the lifted sequence approaches its cluster set, and its values tend to ℓ, so the common neighborhood and gap threshold apply uniformly.

If some value equals ℓ, (39) makes all later steps zero. Otherwise set $e_k = F(r_k) - \ell > 0$ and $\varphi(s) = Cs^{1-\theta}/(1-\theta)$. This function is concave, continuous at zero, and satisfies $\varphi'(e_k)W(r_k) \geq 1$. Consequently

$$\varphi(e_k) - \varphi(e_{k+1}) \geq \varphi'(e_k)(e_k - e_{k+1}) \geq \gamma\|d_k\|_2. \tag{54}$$

Telescoping proves (35), including the finite prefix. Thus r_k is Cauchy and converges to one $\bar{r} \in B$. Nonisolated boundary cluster sets have been handled by the uniform compact-set argument, not assumed absent.

Summing the exact recurrence and using continuity of the gradient gives

$$\frac{q_k}{k} \longrightarrow -\alpha\nabla F(\bar{r}). \tag{55}$$

If $\bar{r}$ is interior, q_k stays bounded and hence $\nabla F(\bar{r}) = 0$. If $\bar{r}$ lies on the sphere, $\|q_k\|_2 \to \infty$ and $q_k/\|q_k\|_2 \to \bar{r}$. Equation (55) then gives $\nabla F(\bar{r}) = -\lambda\bar{r}$ with $\lambda \geq 0$, treating a zero limiting gradient separately. Therefore $\langle \nabla F(\bar{r}), s - \bar{r}\rangle = \lambda(1 - \langle \bar{r}, s\rangle) \geq 0$ for $s \in B$. This proves Theorem 2.

7.4 Complex Hermitian two-by-two density matrices

For $d = 3$, write $r = (u, v, w)$ and define

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \tag{56}$$
$$S(r) = u\sigma_z + v\sigma_x + w\sigma_y, \qquad X(r) = \tfrac{1}{2}(I + S(r)).$$

The eigenvalues of $X(r)$ are $(1 \pm \|r\|_2)/2$. Thus the three-dimensional ball parametrizes all complex Hermitian positive semidefinite trace-one two-by-two matrices; its interior corresponds to positive definite matrices. For the real polynomial extension

$$f_E(X) = F(X_{11} - X_{22}, 2\,\mathrm{Re}\,X_{12}, -2\,\mathrm{Im}\,X_{12}), \tag{57}$$

the real Frobenius gradient on Hermitian matrices is

$$G(X(r)) = F_u(r)\sigma_z + F_v(r)\sigma_x + F_w(r)\sigma_y. \tag{58}$$

Indeed, differentiation in a Hermitian direction verifies this expression with the pairing $\mathrm{Tr}(GH)$, which is real. In particular,

$$\mathrm{Tr}(G(X(r))(X(s) - X(r))) = \langle \nabla F(r), s - r \rangle.$$

Matrix entropy restricts to h_E, and the Bregman divergences agree on the trace slice.

The Pauli identities imply $S(r)^2 = \|r\|_2^2 I$. With $q = \nabla h_E(r)$,

$$\log X(r) = \frac{1}{2}\log\frac{1 - \|r\|_2^2}{4}\, I + S(q), \qquad e^{S(q)} = \cosh\|q\|_2\, I + \frac{\sinh\|q\|_2}{\|q\|_2} S(q), \tag{59}$$

where the zero value is interpreted continuously. Hence (38) is precisely the normalized matrix-exponential update

$$X_{k+1} = \frac{\exp(\log X_k - \alpha G(X_k))}{\mathrm{Tr}\exp(\log X_k - \alpha G(X_k))}. \tag{60}$$

No commutativity or simultaneous diagonalization is used. The norm identities

$$\|X(r) - X(s)\|_F^2 = \tfrac{1}{2}\|r - s\|_2^2, \qquad \|G(X(r)) - G(X(s))\|_F^2 = 2\|\nabla F(r) - \nabla F(s)\|_2^2 \tag{61}$$

show that $2L_E$ is a Frobenius Lipschitz bound on the trace slice and that Theorem 2 gives finite total Frobenius length. The update uses the same α; the kernel's strong-convexity constant in that norm is two.

The limiting matrix satisfies the ordinary variational inequality against every Hermitian positive semidefinite trace-one competitor. If its rank is one, the normal sign can also be written explicitly: $\nabla F(\bar{r}) = -\lambda \bar{r}$ gives the positive semidefinite slack $\lambda(I - S(\bar{r}))$, whose product with $\bar{X}$ is zero. This covers the full complex two-by-two class, but gives no conclusion for larger matrices.

Matrix-exponentiated-gradient methods themselves are classical [6]. Existing convex low-rank implementations [7] address a different objective class, and the quantum common-interest result of Lin et al. [8, Section 2.3 and Theorem 5.6] uses a linear sandwich update and a connected fixed-point accumulation set. Its stated analysis excludes exponential matrix multiplicative weights. Here the additional work is the exact boundary estimate (43); the analytic gradient inequality is used only after that estimate is verified. These comparisons do not assert exhaustive priority.

7.5 A noncommuting rank-one limit

The following $d = 2$ example embeds into the complex two-by-two setting by extending F independently of w and initializing $w = 0$. The exact dual update preserves this real slice.

Example 2. For

$$F(u, v) = (1 - u^2 - v^2)(1 + u/4), \qquad \alpha = \tfrac{1}{8}, \qquad r_0 = (0, \tfrac{1}{2}), \tag{62}$$

$L_E = 7/2$ is valid. The orbit has finite length and converges to a circle point, so its matrix sequence converges to a rank-one matrix. Every point of the circle is a global minimizer and a KKT point. Nevertheless X_0 and X_1 do not commute.

Proof. With $\rho = \sqrt{u^2 + v^2}$, the Hessian is

$$\nabla^2 F = \begin{pmatrix} -2 - 3u/2 & -v/2 \\ -v/2 & -2 - u/2 \end{pmatrix}. \tag{63}$$

Its eigenvalues $-2 - u \pm \rho/2$ lie in $[-7/2, -1/2]$ on the disk. Thus the gradient bound holds and F is strictly concave. An interior stationary point would maximize F on the disk, with value at least $F(0) = 1$. Since $F(r_0) = 3/4$ and the values decrease, the limit from Theorem 2 cannot be interior. The factor $1 + u/4$ is positive on the disk, so precisely the entire circle minimizes F with value zero; there $\nabla F(r) = -2(1 + u/4)r$, giving the KKT condition.

At the prescribed initial point, $g_0 = (3/16, -1)$ and

$$q_0 = (0, \operatorname{atanh}(1/2)), \qquad q_1 = (-3/128, \operatorname{atanh}(1/2) + 1/8). \tag{64}$$

The two vectors are not parallel. Since r_i is a nonzero positive multiple of q_i, the associated Pauli matrices, and hence X_0 and X_1, have a nonzero commutator. Finite length follows from Theorem 2. ∎

8 Selection for globally convex matrix polynomials

The nonconvex result above is restricted to two-by-two matrices. Under global convexity of the objective, a polynomial argument gives initial-entropy selection in every finite matrix size.

Let $n \geq 2$ and let $\mathcal{H}_n$ denote the real Hilbert space of complex Hermitian $n \times n$ matrices, with Frobenius pairing $\langle X, Y \rangle = \operatorname{Tr}(XY)$. Set

$$\mathcal{D}_n = \{X \in \mathcal{H}_n : X \succeq 0,\ \operatorname{Tr} X = 1\}. \tag{65}$$

Let $f : \mathcal{H}_n \to \mathbb{R}$ be a real polynomial that is convex on the *entire* space $\mathcal{H}_n$. Choose any $L > 0$ such that

$$\|\nabla f(X) - \nabla f(Y)\|_F \leq L\|X - Y\|_F \qquad (X, Y \in \mathcal{D}_n). \tag{66}$$

Such a bound exists on the compact density domain. Global Lipschitz continuity of the gradient is not required. Fix $0 < \alpha < 1/L$ and any positive definite $X_0 \in \mathcal{D}_n$, and run

$$X_{k+1} = \frac{\exp\bigl(\log X_k - \alpha \nabla f(X_k)\bigr)}{\operatorname{Tr}\exp\bigl(\log X_k - \alpha \nabla f(X_k)\bigr)}. \tag{67}$$

For $Y \succeq 0$ and $X \succ 0$ in $\mathcal{D}_n$, define

$$D(Y\|X) = \operatorname{Tr}(Y \log Y) - \operatorname{Tr}(Y \log X), \qquad 0 \log 0 = 0. \tag{68}$$

Theorem 3. *The set $\mathcal{S} = \arg\min_{\mathcal{D}_n} f$ is nonempty, compact and convex, and there is a unique*

$$X^\dagger = \arg\min_{Y \in \mathcal{S}} D(Y\|X_0). \tag{69}$$

Every finite iterate (67) *is positive definite. Its objective values are nonincreasing, its squared Frobenius steps are summable, and*

$$D(X^\dagger\|X_k) \longrightarrow 0, \qquad \|X_k - X^\dagger\|_F \longrightarrow 0. \tag{70}$$

These conclusions allow nonunique minimizers, singular $X^\dagger$ and noncommuting matrices. No boundedness assumption is imposed on the logarithms or accumulated gradients. No finite-length or state-rate conclusion is asserted.

The ambient convexity assumption is part of the theorem; convexity only on $\mathcal{D}_n$ is insufficient for the argument below. The theorem makes no assertion for arbitrary smooth nonpolynomial losses. Constant and linear polynomials are included by choosing a positive Lipschitz upper bound in (66).

A quadratic instance is

$$f_Q(X) = \tfrac{1}{2}\|AX - b\|_2^2, \qquad A : \mathcal{H}_n \to \mathbb{R}^m \text{ real linear}, \quad b \in \mathbb{R}^m. \tag{71}$$

For $A \neq 0$, one may take $L = \|A\|^2$ in the Frobenius-to-Euclidean norm. The data need not satisfy $b \in A(\mathcal{D}_n)$. If $A = 0$, the orbit is constant for any positive step. For feasible quadratic interpolation, entropy selection is already an instance of Jin's dual-gradient theorem [9, Assumption 1 and Theorem 3.1(i)]. General linearized-Bregman selection results, such as Dai, Deng and Zhang [10, Theorem 4.2], treat finite full-domain penalties. Here the closed density constraint and possibly singular solution are retained. The spectral failure of boundary reciprocity is an established issue [11, Remark 4]; the proof below establishes the displayed entropy limit directly. The comparison with [10] concerns the stated public version, not an uninspected final journal theorem.

8.1 Entropy descent and summable objective gaps

Write $h(X) = \operatorname{Tr}(X \log X)$. For a positive definite density matrix with eigenvalues λ_i, differentiation in an eigenbasis gives

$$D^2 h(X)[U,U] = \sum_{i,j} |U_{ij}|^2 \int_0^\infty \frac{dt}{(\lambda_i + t)(\lambda_j + t)} \geq \|U\|_F^2, \tag{72}$$

since $0 < \lambda_i \le 1$. Integrating along a segment, then approximating singular endpoints by positive definite densities, proves strong convexity of h on $\mathcal{D}_n$ and

$$D(Y\|X) \ge \tfrac{1}{2}\|Y - X\|_F^2 \qquad (Y \in \mathcal{D}_n,\ X \succ 0). \tag{73}$$

The function

$$H_0(Y) = D(Y\|X_0) = h(Y) - \langle \log X_0, Y\rangle \tag{74}$$

is finite, continuous and strictly convex on the compact density domain. Thus it has exactly one minimizer on the nonempty compact convex set $\mathcal{S}$.

Put $g_k = \nabla f(X_k)$ and $d_k = X_{k+1} - X_k$. A finite Hermitian exponential is positive definite, so the recurrence stays in the interior. Its exact logarithmic identity is

$$\log X_{k+1} = \log X_k - \alpha g_k - \gamma_k I, \qquad \gamma_k = \log \operatorname{Tr} \exp(\log X_k - \alpha g_k). \tag{75}$$

For any $Y \in \mathcal{D}_n$, direct expansion gives

$$\alpha\langle g_k, X_{k+1} - Y\rangle = D(Y\|X_k) - D(Y\|X_{k+1}) - D(X_{k+1}\|X_k). \tag{76}$$

The scalar identity term cancels by the trace constraint. Taking $Y = X_k$, and using smoothness only along the segment between the two iterates, yields

$$f(X_k) - f(X_{k+1}) \ge (\alpha^{-1} - L/2)\|d_k\|_F^2. \tag{77}$$

This proves objective decrease and square-step summability.

Let $f_* = \min_{\mathcal{D}_n} f$, $a_k = f(X_k) - f_*$, $\mathcal{E}_k = D(X^\dagger\|X_k)$ and $B_k = D(X_{k+1}\|X_k)$. Convexity at X_k and smoothness give $a_{k+1} \le \langle g_k, X_{k+1} - X^\dagger\rangle + (L/2)\|d_k\|_F^2$. Substituting in (76) gives

$$\mathcal{E}_k - \mathcal{E}_{k+1} \ge \alpha a_{k+1} + (1 - \alpha L)B_k. \tag{78}$$

Consequently (a_k) is nonnegative, nonincreasing and summable. In particular

$$k a_k \longrightarrow 0, \qquad \operatorname{dist}(X_k, \mathcal{S}) \longrightarrow 0, \qquad \liminf_k H_0(X_k) \ge H_0(X^\dagger). \tag{79}$$

For the first assertion, bound $(k/2)a_k$ by the sum from $\lfloor k/2\rfloor$ to k. The remaining assertions follow by compactness and continuity, since every cluster point minimizes f.

8.2 A global polynomial pairing estimate

The next elementary estimate supplies the extra control used for selection.

Lemma 4. *Let r be a globally convex real polynomial of degree at most $d \ge 1$ on a finite-dimensional real Hilbert space, with $r \ge 0$ and $r(0) = 0$. Then*

$$|\langle \nabla r(u), v\rangle| \le C_d\big(r(u) + r(v)\big), \qquad C_d = \frac{(6d)^d}{2d!}, \qquad u, v \text{ arbitrary.} \tag{80}$$

Proof. For fixed v, the polynomial $q(t) = r(tv)$ has degree at most d and satisfies $0 \le q(t) \le q(1)$ on $[0,1]$. Interpolate at $t_j = j/d$, $0 \le j \le d$. At either $t = 2$ or $t = -2$, every numerator factor of a Lagrange basis polynomial has absolute value at most 3, whereas its denominator has absolute value $d^{-d} j!(d-j)!$. Thus

$$r(\pm 2v) \le (3d)^d \sum_{j=0}^{d} \frac{r(v)}{j!(d-j)!} = 2C_d r(v). \tag{81}$$

This includes directions with $r(v) = 0$ and the zero polynomial. Convexity now gives $r(u \pm v) \le \frac{1}{2}[r(2u) + r(\pm 2v)] \le C_d(r(u) + r(v))$. The supporting inequalities $r(u \pm v) \ge r(u) \pm \langle \nabla r(u), v \rangle$, together with $r(u) \ge 0$, prove (80). ∎

Set $G = \nabla f(X^\dagger)$ and define the ambient polynomial

$$r(Z) = f(X^\dagger + Z) - f_* - \langle G, Z \rangle. \tag{82}$$

Global convexity makes r globally nonnegative and convex, with $r(0) = 0$. Constrained optimality of $X^\dagger$ gives, for $X \in \mathcal{D}_n$,

$$0 \le \langle G, X - X^\dagger \rangle \le f(X) - f_*, \qquad 0 \le r(X - X^\dagger) \le f(X) - f_*. \tag{83}$$

For $Y \in \mathcal{S}$ both terms vanish. Convexity implies that $r(t(Y - X^\dagger)) = 0$ on $[0,1]$, and polynomiality extends this identity to every real t. At any ambient point Z, the supporting inequality therefore gives $0 \ge r(Z) + \langle \nabla r(Z), t(Y - X^\dagger) - Z \rangle$ for all real t. Its linear coefficient must be zero. Hence

$$\langle \nabla r(Z), Y - X^\dagger \rangle = 0 \qquad (Z \in \mathcal{H}_n,\ Y \in \mathcal{S}). \tag{84}$$

Both global convexity and polynomiality are used at this point, outside the feasible density domain.

8.3 Vanishing accumulated pairing and entropy selection

Write $z_k = X_k - X^\dagger$ and consider the actual accumulated pairing

$$T_k = \langle \sum_{j<k} g_j, z_k \rangle = k\langle G, z_k \rangle + \sum_{j<k} \langle \nabla r(z_j), z_k \rangle. \tag{85}$$

The first term tends to zero by (79) and (83). Fix N. Each of the finitely many terms with $j < N$ tends to zero by (84) and $\operatorname{dist}(X_k, \mathcal{S}) \to 0$. For the remaining terms Lemma 4 yields

$$\left| \sum_{j=N}^{k-1} \langle \nabla r(z_j), z_k \rangle \right| \le C_d \left(k a_k + \sum_{j=N}^{\infty} a_j \right), \tag{86}$$

where any degree bound $d \ge 1$ for r is allowed. First let $k \to \infty$, then $N \to \infty$. Summability proves $T_k \to 0$.

Finally sum (75) and pair it with $X^\dagger$ and X_k. Eliminating the scalar normalizer gives the exact identity

$$D(X^\dagger \| X_k) = H_0(X^\dagger) - H_0(X_k) - \alpha T_k. \tag{87}$$

Its left side is nonnegative, while (79) and $T_k \to 0$ make its upper limit nonpositive. It therefore tends to zero. Inequality (73) proves whole-state convergence, completing Theorem 3. Only entropy with the fixed positive definite second argument X_0 was treated as a continuous function on the closed density domain; no boundary reciprocity or bounded-score assumption was used.

8.4 An inconsistent three-level example

Let E_{33} have only its $(3,3)$ entry equal to one. Take

$$A(X) = X_{33}, \quad b = -1, \quad \alpha = \tfrac{1}{2}, \qquad Q_0 = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad X_0 = \frac{e^{Q_0}}{\operatorname{Tr} e^{Q_0}}. \tag{88}$$

Here $\|A\| = 1$, while $A(\mathcal{D}_3) = [0,1]$. The minimum objective value is $1/2$, attained on the entire face $X_{33} = 0$. Positive semidefiniteness makes this the density-matrix face supported on the first two coordinates. On that face the compressed initial logarithm differs from $\operatorname{diag}(1,0)$ by a scalar identity, so entropy minimization gives

$$X^\dagger = \operatorname{diag}\left(\frac{e}{e+1}, \frac{1}{e+1}, 0\right). \tag{89}$$

Theorem 3 therefore proves convergence to this rank-two point in a nonisolated solution face.

For the actual orbit one can also write

$$X_k = \frac{\exp(Q_0 - \tau_k E_{33})}{\operatorname{Tr}\exp(Q_0 - \tau_k E_{33})}, \qquad \tau_0 = 0, \quad \tau_{k+1} = \tau_k + \tfrac{1}{2}((X_k)_{33} + 1). \tag{90}$$

In particular $k/2 \le \tau_k \le k$. The first score increment is positive and $[Q_0, E_{33}] \neq 0$, so $\log X_0$ and $\log X_1$ do not commute. The same is true of X_0 and X_1. This is one fixed initialized trajectory; it makes no separate rate or finite-length assertion.

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